AquaDissect Defense FAQ

Common panel questions and suggested answers

Rehearse these questions as a team. This guide uses the 4 November ethics revision and software checks available on 7 November 2026. Update pending answers from verified records before the defense.

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Project and scope

01 What is AquaDissect?

AquaDissect is a browser-based pre-laboratory tool for MB 304L Marine Zoology II Vertebrates Laboratory. Students preview Rabbitfish anatomy, safety and procedure before supervised dissection. The proposed study setting is Southern Leyte State University Bontoc Campus.

02 What problem are you addressing?

Students need time to revisit structures and procedures before handling a specimen. Limited specimens, laboratory time, and opportunities for repeated practice motivate the project. The app offers guided visuals and labels that students can review at their own pace.

03 Why focus on Rabbitfish if the working title mentions selected organisms?

The 4 November draft retains *AquaDissect: Guided Dissection Learning and Anatomical Identification System for Selected Marine Organisms* as its working title. The current study covers Rabbitfish (danggit). Faculty still needs to confirm the specimen identity. The draft package uses Goldspotted Rabbitfish (*Siganus guttatus*).

04 Does AquaDissect replace the physical dissection?

The app prepares students for the physical laboratory. Students preview anatomy and safety and practice recognizing structures in the browser. Faculty supervise the actual specimen handling, tool control, and practical observation, where students apply what they prepared.

05 Who is responsible for the study and who is the client?

The BSIT proponents manage the protocol and ethics submission and conduct the approved study. Marine Biology personnel are the clients and subject-matter representatives. The team will document who reviews the material, supervises the lab, recruits participants, and evaluates the research.

System and access

06 What can you demonstrate in the application?

With a fictional account, we can show roster-based sign-in, orientation, and section readiness. In the first-login workflow, a student changes the temporary password before opening the course. The Rabbitfish viewer uses layered 2.5D images, a depth control, selectable structures, and formative questions.

07 How do the four roles control what students can open?

Students work in their assigned sections. Instructors manage activities and progress for their sections. The Program Head can oversee sections and record independent faculty decisions. Administrators manage access and publish approved versions. Students receive an assigned published lesson after satisfying any applicable Safety & Tools requirement.

08 Why use a web application, and how is it built?

Students open their assigned material in a supported browser. Laravel handles accounts, permissions, and progress records. Lesson packages hold images, labels, steps, and questions in versions so faculty can review the exact material being released.

09 How does offline access work?

Learners download permitted material while online. Supported formative work can be saved on the device, then synced under the original account and section. We check server confirmation after reconnection. Formal research tests in the application require an online session.

10 Can the on-screen ruler measure the actual specimen?

The ruler gives a relative image distance by default. Physical units require a reviewed calibration that matches the exact image and permitted measurement scope. A calibrated illustration is labeled as an estimate. Actual specimen measurements come from the approved physical laboratory procedure.

11 Are the Rabbitfish clips AI-generated, and who checks them?

The four preparation clips in Rabbitfish v1.2.0 are identified as AI-generated teaching media in the package records. They are available for faculty review. Marine Biology reviewers must verify specimen identity, organ placement, labels, procedure, and safe handling before the relevant version is approved for student use.

Research design

12 What is your general objective?

We aim to develop and evaluate a syllabus-aligned Rabbitfish pre-laboratory activity for MB 304L. We plan to assess knowledge and perceived preparedness before and after use, describe post-use practical performance, and obtain separate evidence on software quality and curriculum alignment.

13 What learning principles guide the activity?

Multimedia learning informs the short steps and the placement of labels near the relevant structures. Experiential learning informs the sequence. Students preview and reflect on the procedure, then apply that preparation during supervised physical work.

14 What research design will you use, and what can it show?

We propose a developmental-evaluative design with a one-group pretest-posttest component. It lets us compare the same students' scores before and after the activity. Concurrent instruction, additional practice, or repeated testing could contribute to any change, so we will report the observed difference in this group and its limitations.

15 Who can participate, and who is excluded?

Eligible students are enrolled in confirmed MB 304L sections, are at least 18, and give informed consent. Students outside the confirmed setting, minors, people who decline, and demonstration accounts are excluded from the research. Qualified faculty and technical reviewers have separate roles and evaluation instruments.

16 What does total enumeration mean for your sample?

We will invite every eligible adult in the confirmed sections. Participation is voluntary. We will report enrolled, eligible, invited, consenting, and valid paired counts separately. The final participant count comes from verified recruitment and study records.

17 What will participants do, and in what order?

After consent, we assign a study code and collect the knowledge and preparedness baseline before exposure. Participants complete the approved virtual preparation, then the posttest and preparedness scale at the approved interval. We will observe practical performance during supervised dissection and collect separate software-experience responses.

Measures and analysis

18 What instruments are ready for review?

The draft pack contains 25 candidate knowledge questions, a proposed Table of Specifications, pretest and posttest copies, and a restricted reviewer key. The current same-form approach needs approval or validated alternatives. The pack also includes a six-item preparedness scale, a four-point practical rubric, and separate quality and content-review instruments.

19 How will you validate the instruments?

We intend to obtain review from at least three qualified validators, subject to institutional requirements. They will check alignment with the syllabus and local lab procedure, item wording, answer keys, scoring, and rubric descriptors. We plan to pilot outside the final respondent group where feasible and approved. Validation records and any reliability estimates still need evidence.

20 How will you analyze the before-and-after scores?

We will summarize knowledge and preparedness separately using valid linked pairs. A paired t-test is one option when the differences are approximately normal and its other assumptions hold. We will agree on the alternative in advance and report the pair count, estimated change, effect size, and uncertainty where justified. Missing outcomes stay separate from zero scores.

21 How do you distinguish confidence from practical performance?

The preparedness scale records what participants say about their readiness. The practical rubric records observed behavior in preparation and safety, tool handling, procedure, organ identification, specimen handling, and cleanup. The proposal measures practical performance after use, so we will report the observed level at that point.

22 What will ISO/IEC 25010 tell you?

The draft uses selected ISO/IEC 25010:2023 product-quality criteria. Students judge their own experience, while qualified experts assess criteria within their expertise. Report the evaluator groups separately. Learning change is assessed through the research measures described above. ISO reference

Ethics and privacy

23 How did you address the ethics committee’s comments?

The revised set includes an adult information and consent draft, separate permission requests for the BSMB Program Head and MB 304L instructor, and candidate research instruments. Before recruitment or a research test run, obtain ethical clearance, both signed permissions, and approved participant materials and validated instruments.

24 What will participants be told before they consent?

The approved information sheet will describe what participants will do, the time required, risks, what we collect, who can access records, and how records will be retained and disposed of. Adults can ask questions, skip research items, or stop. The team must confirm the official contacts.

25 How will you prevent pressure from a student’s instructor?

The proposal uses a neutral recruiter and private consent. Students who decline continue ordinary course activities. We will hold research scores and ratings separately from grades and use an approved separate collection method if course dashboards cannot enforce access restrictions. Disclose any instructor role in research observation.

26 How will you protect the data, and how long will it be kept?

We plan to use participant codes and keep consent and the linkage key in separate, restricted storage. We will report grouped findings. The institutional custodian must confirm who can access the files, the retention period and starting event for each record type, and the disposal and backup arrangements.

27 What happens if someone withdraws from the study?

A participant can stop without penalty and request removal of linked research responses before the approved cutoff. Once responses are irreversibly anonymized and combined, we may be unable to identify them. The team will put the confirmed cutoff and withdrawal contact in the approved consent form.

28 How will you manage risk during the practical activity?

Qualified faculty will supervise the laboratory and can stop unsafe work. The team must confirm the specimen and materials, required protective equipment, local handling rules, and the response and referral plan. For research participation, also plan for fatigue, test anxiety, and device or connectivity problems.

Evidence and demo

29 What can you conclude at this point?

We can show the implemented prototype and the workflows exercised in engineering tests. The reviewed materials contain no verified participant findings. Any claim about learning or preparedness must come from approved data collection, analysis of validated measures, and an interpretation that fits the study design.

30 Which technical problems did you fix and verify?

Recent fixes keep Safety & Tools progress in the selected section and let older imported test forms open correctly. We checked existing offline downloads after updating the app. The release passed 218 automated application tests and five recovery checks. Browser verification used fictional accounts.

31 Are the latest fixes live?

The source updates were verified on GitHub. On 7 November 2026 the live sign-in page timed out and GitHub reported a host connection failure during deployment. Verify the live site and release before presenting the fixes as available there. GitHub release changes

32 What should you show if the live site is unavailable?

Use a rehearsed local build with fictional accounts and sections. Show sign-in and readiness, the appropriate staff preview of material under review, and saved progress in an isolated demo. Keep a short recording or screenshots available if the local demo also fails. State which environment and content version the panel is seeing.

33 How do you explain your development method and contributions?

The project uses Agile software development with an adapted Scrum framework. Show the actual backlog, dated increments, review notes, corrections, and contribution records. Explain each person’s documented role. The developmental-evaluative protocol governs research with participants and has its own permissions and evidence.

Defense checklist

Confirm these details with the proponents and adviser before rehearsal. Bring records the panel can inspect and use the pending status where evidence is still being gathered.

- [ ] **Title and study setting**

  Agree on the official title, Rabbitfish scope, course wording, eligible sections, schedule, and expected activity duration.

- [ ] **Clearance and signed permissions**

  Bring the current ethics decision, signed Program Head and instructor permissions, approved adult consent, and the private arrangement for nonparticipants.

- [ ] **Faculty review and validated instruments**

  Bring the reviewed species identity, syllabus and lab procedure mapping, approved versions, validated test forms and scoring, and the practical rubric.

- [ ] **Recruitment, data access, and analysis**

  Confirm the neutral recruiter, rater and data custodian, retention and withdrawal rules, eligible and paired counts, and the adviser-approved analysis.

- [ ] **Demo and results evidence**

  Confirm the build, demo accounts, and fallback recording. Bring validated results and actual analysis outputs if available. Keep reviewer keys and identifiable records private.

References for rehearsal

AquaDissect full proposal, research instruments, adult consent, and ethics response and completion record. Ethics revision draft 2.0, 4 November 2026.

AquaDissect Rabbitfish working manuscript, Draft 1.2.6, and Project 2 Study Alignment. Use Chapter IV system figures for the development demonstration.

For role-specific controls and the local demo, see the AquaDissect User Manual.